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VALVE STRUCTURE FOR HYDROGEN TANK

Abstract

The valve structure of the hydrogen tank includes a valve body mounted on a base of the hydrogen tank and having a first communication passage communicating with the inside of the hydrogen tank, and a fusible plug valve mounted on the valve body via a spacer. The spacer has a second communication passage communicating with the first communication passage and a discharge port provided on an outer wall of the spacer. The second communication passage and the discharge port are shut off by the valve body when the fusible plug valve is closed, and communicate with each other when the fusible plug valve is opened. The discharge port is provided on an outer wall of the spacer other than an outer wall that contacts the valve body and an outer wall that contacts the fusible plug valve.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-030277 filed on Feb. 29, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a valve structure for a hydrogen tank.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2022-88966 (JP 2022-88966 A) discloses a fuel cell electric vehicle including a hydrogen tank and a hydrogen pipe connecting the hydrogen tank and a fuel cell and provided with a discharge port that discharges hydrogen to the outside. In this fuel cell electric vehicle, the hydrogen pipe is provided with a heat-actuated overpressure suppression device and a pressure-actuated overpressure suppression device. With this configuration, hydrogen flowing through the hydrogen pipe is discharged from the discharge port when the internal temperature of the hydrogen tank is equal to or higher than a predetermined temperature or the internal pressure of the hydrogen tank is equal to or higher than a predetermined pressure, maintaining the safety of the hydrogen tank.

SUMMARY

[0004] In the fuel cell electric vehicle described in JP 2022-88966 A, however, it is assumed that a hydrogen pipe is used, complicating the valve structure. Additionally, the discharge port is provided in the hydrogen pipe connecting the hydrogen tank and the fuel cell, and therefore there is a problem that the position of the discharge port is not freely changeable.

[0005] The present disclosure has been made to address such technical issues, and an object thereof is to provide a valve structure for a hydrogen tank capable of changing the position of a discharge port while suppressing the complication of the structure.

[0006] An aspect of the present disclosure provides a valve structure for a hydrogen tank, including: [0007] a valve body mounted to a metal fitting of the hydrogen tank and having a first communication passage communicating with an interior of the hydrogen tank; and [0008] a fusible plug valve attached to the valve body via a spacer, in which: [0009] the spacer has a second communication passage communicating with the first communication passage, and a discharge port provided in an outer wall of the spacer; [0010] the second communication passage and the discharge port are blocked by a valve element of the fusible plug valve when the fusible plug valve is closed, and communicate with each other when the fusible plug valve is opened; and [0011] the discharge port is provided in a part of the outer wall of the spacer other than the outer wall in contact with the valve body and other than the outer wall in contact with the fusible plug valve. [0012] In the valve structure for a hydrogen tank according to the present disclosure, the fusible plug valve is attached to the valve body via a spacer, simplifying the structure compared to the conventional structure having a hydrogen pipe. In addition, the fusible plug valve is not directly attached to the valve body, and is attached to the valve body via the spacer, enabling the discharge port for a hydrogen gas to be provided in the spacer rather than being provided in the valve body. Thus, the number of locations where the discharge port can be arranged increases. Additionally, the discharge port is provided in a part of the outer wall of the spacer other than the outer wall in contact with the valve body and other than the outer wall in contact with the fusible plug valve. Therefore, the position of the discharge port is freely changeable in a part of the outer wall of the

spacer other than the outer wall in contact with the valve body and other than the outer wall in contact with the fusible plug valve. Thus, it is possible to adjust the position of the discharge port so as not to interfere with peripheral components according to the arrangement relationship of the peripheral components of the valve structure. As a result, it is possible to implement a valve structure for a hydrogen tank capable of changing the position of a discharge port while suppressing the complication of the structure.

[0013] In the valve structure according to the present disclosure, [0014] the fusible plug valve may include a pressing member that presses the valve element in a valve closing direction, and a fusible member juxtaposed to the pressing member; and [0015] the discharge port may be disposed on a side opposite to the fusible member when viewed in a direction orthogonal to a direction in which the pressing member and the fusible member are juxtaposed.

With this configuration, it is possible to suppress the influence of hydrogen discharge on the fusible plug valve.

[0016] According to the present disclosure, it is possible to implement a valve structure for a hydrogen tank capable of changing the position of a discharge port while suppressing the complication of the structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0018] FIG. 1 is a perspective view illustrating a valve structure of a hydrogen tank according to an embodiment;

[0019] FIG. 2 is a side view illustrating a valve structure of the hydrogen tank according to the embodiment;

[0020] FIG. 3 is a cross-sectional view taken along A-A line of FIG. 2;

[0021] FIG. 4 is a cross-sectional view taken along B-B line of FIG. 2;

[0022] FIG. 5 is a cross-sectional view showing the valve opening of the fusible plug valve; and

[0023] FIG. 6 is a cross-sectional view illustrating a valve opening state of the fusible plug valve.

DETAILED DESCRIPTION OF EMBODIMENTS

[0024] Hereinafter, an embodiment of a valve structure of a hydrogen tank according to the present disclosure will be described with reference to the drawings. In the description of the drawings, the same elements are denoted by the same reference numerals, and redundant description thereof will be omitted. It should also be noted that the valve structure of the hydrogen tank is not meant to include a base as well, although the base of the hydrogen tank is also depicted in the drawings in order to make the valve structure of the hydrogen tank easier to understand.

[0025] FIG. 1 is a perspective view showing a valve structure of a hydrogen tank according to an embodiment, and FIG. 2 is a side view showing a valve structure of a hydrogen tank according to an embodiment. FIG. 3 is a cross-sectional view taken along A-A line of FIG. 2, and FIG. 4 is a cross-sectional view taken along B-B line of FIG. 2. The valve structure 1 (hereinafter, abbreviated as “valve structure 1”) of the hydrogen tank according to the present embodiment is used by being attached to a base 100 of a hydrogen tank (not shown).

[0026] Here, the structure of the hydrogen tank will be briefly described. The hydrogen tank is a hollow container that stores high-pressure hydrogen. The hydrogen tank includes, for example, a substantially cylindrical liner having both ends rounded in a dome shape, a fiber-reinforced resin layer (also referred to as a reinforcing layer) formed so as to cover the outer peripheral surface of the liner, and a base 100 attached to one end portion of the hydrogen tank.

[0027] The base **100** is formed by processing a metal material such as stainless steel or an aluminum alloy into a predetermined shape. The base **100** has a substantially cylindrical base body portion **101** extending along the axial direction of the hydrogen tank, and a flange portion **102** integrally formed with the base body portion **101** and protruding in the radial direction of the hydrogen tank. The base **100** having such a structure is fixed to the end portion of the hydrogen tank by sandwiching the flange portion **102** between the liner and the reinforcing layer. Further, a female screw portion for screwing with the valve structure **1** is formed on the inner peripheral wall of the base body portion **101**.

[0028] The valve structure **1** mainly includes a valve body **2** that is attached to the base **100** of the hydrogen tank and has a first communication passage **20** that communicates with the inside of the hydrogen tank, and a fusible plug valve **4** that is attached to the valve body **2** via a spacer **3**. The valve body **2** is formed of a metal material such as stainless steel or an aluminum alloy, and has an insertion portion **21** inserted into the inside of the base **100**, and a protruding portion **22** integrally formed with the insertion portion **21** and protruding from the base **100**.

[0029] The insertion portion **21** has a cylindrical shape extending along the axial direction of the hydrogen tank, and a male screw portion that is threadedly engaged with a female screw portion formed on an inner peripheral wall of the base **100** is formed on an outer peripheral wall thereof. The protruding portion **22** has a substantially rectangular shape and extends in the radial direction of the insertion portion **21** (that is, in the radial direction of the hydrogen tank) so as to have a T-shaped cross section with respect to the insertion portion **21**.

[0030] **10** As illustrated in FIGS. **3** and **4**, the first communication passage **20** is provided inside the insertion portion **21** and the protruding portion **22**. The first communication passage **20** has an axial portion **201** and a radial portion **202**. The axial portion **201** is formed along the axial direction of the insertion portion **21** (that is, the axial direction of the hydrogen tank) over the entire length of the insertion portion **21** and the protruding portion **22**. The radial portion **202** communicates with the axial portion **201** and extends in the radial direction of the insertion portion **21** toward the spacer **3**.

[0031] The spacer **3** is formed in a substantially block shape by a metal material such as stainless steel or an aluminum alloy, and is fixed to the side wall of the protruding portion **22** by a bolt **5**. Here, the side wall refers to an outer wall extending along the axial direction of the insertion portion **21**. The spacer **3** has a second communication passage **30** communicating with the first communication passage **20**, and a discharge port **31** provided on the outer wall of the spacer **3**.

[0032] As illustrated in FIGS. **3** and **4**, the second communication passage **30** includes a first portion **301** and a second portion **302**. The first portion **301** communicates with the radial portion **202** of the first communication passage **20** and extends in the radial direction of the insertion portion **21**. The second portion **302** extends in a direction orthogonal to the extending direction of the first portion **301** so as to communicate the first portion **301** and the discharge port **31**. The first portion **301** is formed so that its diameter is slightly smaller than the diameter of the radial portion **202** of the first communication passage **20**, and the second portion **302** is formed so that its diameter increases toward the discharge port **31** (see FIG. **4**).

[0033] The second communication passage **30** and the discharge port **31** are cut off by the valve body **42** of the fusible plug valve **4** when the valve of the fusible plug valve **4** is closed, and are formed so as to communicate with each other when the valve of the fusible plug valve **4** is opened.

[0034] the fusible plug valve **4** is disposed on the opposite side of the valve body **2** with the spacer **3** interposed therebetween, and is fixed to the side wall of the spacer **3** by bolts. the fusible plug valve **4** is a safety valve for opening the valve and discharging the high-pressure hydrogen gas in the hydrogen tank when the predetermined temperature or the predetermined pressure is reached, and is, for example, a thermally actuated safety valve (TPRD, Thermal pressure relief device).

[0035] the fusible plug valve **4** includes a valve housing **41**, a valve body **42**, a pressing member **43**, a fusible member **44** arranged in parallel with the pressing member **43**, and a coil spring **45**.

The valve body **42** blocks communication with the second communication passage **30** and the discharge port **31**. The pressing member **43** presses the valve body **42** in the valve closing direction. The coil spring **45** biases the pressing member **43** toward the fusible member **44**.

[0036] The valve body **42** has a substantially circular rod shape and is formed to be movable in the first portion **301** of the second communication passage **30** by the opening and closing operation of the fusible plug valve **4**. Specifically, one end of the valve body **42** enters the first portion **301** of the second communication passage **30** and closes the first portion **301**, and the other end thereof is in contact with the pressing member **43** via the spherical bearing **46**.

[0037] The pressing member **43** is formed in a cylindrical shape, and is accommodated in the valve housing **41** so that its axial direction is along the longitudinal direction of the valve housing **41**. The coil spring **45** is accommodated in the valve housing **41** in a state of being compressed in the axial direction of the pressing member **43**, and biases the pressing member **43** toward the fusible member **44**.

[0038] The fusible member **44** is formed in a disk shape by a material that melts when the ambient temperature reaches a predetermined temperature, for example, a low-melting-point metal material such as lead or tin. The fusible member **44** is arranged in parallel with the pressing member **43** along the longitudinal direction of the valve housing **41** (that is, the axial direction of the pressing member **43**). The fusible member **44** is accommodated in the valve housing **41** so as to close the open end **411** of the valve housing **41** (see FIG. 4).

[0039] As shown in FIG. 4, in the valve-closed state of the fusible plug valve **4**, the pressing member **43** presses the valve body **42** toward the second communication passage **30** via the bearing **46** that abuts against the valve body. As a result, the valve body **42** is pushed into the first portion **301** of the second communication passage **30**. Therefore, the second communication passage **30** and the discharge port **31** are shut off by the valve body **42**.

[0040] On the other hand, when the ambient temperature of the fusible plug valve **4** reaches a predetermined temperature, as shown in FIGS. 5 and 6, the fusible member **44** melts, and the open end **411** of the valve housing **41** opens. Accordingly, the pressing member **43** moves to the space in which the fusible member **44** is accommodated by the urging force of the coil spring **45**. The valve body **42** is pressed toward the fusible plug valve **4** by the pressure of the high-pressure hydrogen. As a result, the fusible plug valve **4** is opened, and the second communication passage **30** and the discharge port **31** communicate with each other. At this time, the hydrogen gas stored in the hydrogen tank is discharged to the atmosphere via the first communication passage **20**, the second communication passage **30**, and the discharge port **31** (see arrows in FIGS. 5 and 6).

[0041] In the present embodiment, the discharge port **31** is provided on the outer wall of the spacer **3** other than the outer wall that contacts the valve body **2** and the outer wall other than the outer wall that contacts the fusible plug valve **4**. For example, in FIG. 1, in a case where the upper and lower sides of the paper surface are in the vertical direction, the left and right sides of the paper surface are in the front-rear direction, and the direction orthogonal to the upper, lower, and left and right sides of the paper surface is in the horizontal direction, the left-side outer wall of the upper, lower, front-rear, and left-right outer walls of the spacer **3** is in contact with the valve body **2**, and the right-side outer wall is in contact with the fusible plug valve **4**. Therefore, the discharge port **31** can be provided in any of the outer walls excluding the outer walls, that is, the upper outer wall, the lower outer wall, the front outer wall, and the rear outer wall. In this way, the arrangement position of the discharge port **31** can be selected. In other words, the orientation of the discharge port **31** can be adjusted. Therefore, by changing the position of the discharge port **31** according to the arrangement relationship of the peripheral components of the valve structure **1**, in particular, the arrangement state of the components around the spacer **3**, it is possible to prevent the hydrogen gas to be discharged from interfering with the peripheral components.

[0042] Further, when viewed from a direction orthogonal to the direction in which the pressing member **43** and the fusible member **44** are arranged in parallel (that is, the longitudinal direction of

the valve housing **41**), the discharge port **31** is preferably arranged on the side opposite to the fusible member **44**. Specifically, for example, in the case where the left-right direction of the paper surface is the front-rear direction in FIG. **4**, the pressing member **43** and the fusible member **44** are arranged side by side along the front-rear direction (that is, the longitudinal direction of the valve housing **41**). Since the fusible member **44** is located on the rear side of the pressing member **43**, the discharge port **31** is disposed on the front side opposite to the fusible member **44**, that is, on the front side outer wall of the spacer **3**. By disposing the discharge port **31** on the side opposite to the fusible member **44** in this way, it is possible to suppress the influence of the hydrogen discharge on the fusible plug valve **4**.

[0043] In the valve structure **1** configured as described above, since the fusible plug valve **4** is attached to the valve body **2** via the spacer **3**, the structure is simplified as compared with a conventional structure having a hydrogen pipe. Further, the fusible plug valve **4** is not directly attached to the valve body **2**, but is attached to the valve body **2** via the spacer **3**. Therefore, the discharge port **31** for the hydrogen gas can be provided in the spacer **3** without being provided in the valve body **2**, and the number of locations where the discharge port **31** can be disposed increases.

[0044] In addition, the discharge port **31** is provided on an outer wall of the outer wall of the spacer **3** other than the outer wall in contact with the valve body **2** and the outer wall other than the outer wall in contact with the fusible plug valve **4**. Therefore, the position of the discharge port **31** can be arbitrarily adjusted as long as it is the outer wall of the spacer **3** other than the outer wall in contact with the valve body **2** and the outer wall other than the outer wall in contact with the fusible plug valve **4**. Therefore, the position of the discharge port **31** can be changed so as not to interfere with the peripheral component according to the arrangement relationship of the peripheral component of the valve structure **1**. As a result, it is possible to realize the valve structure of the hydrogen tank capable of changing the position of the discharge port **31** while suppressing the complication of the structure.

[0045] Although the embodiment of the disclosure has been described in detail above, the disclosure is not limited to the embodiment described above, and various design changes can be made without departing from the spirit of the disclosure described in the claims.

Claims

1. A valve structure for a hydrogen tank, comprising: a valve body mounted to a metal fitting of the hydrogen tank and having a first communication passage communicating with an interior of the hydrogen tank; and a fusible plug valve attached to the valve body via a spacer, wherein: the spacer has a second communication passage communicating with the first communication passage, and a discharge port provided in an outer wall of the spacer; the second communication passage and the discharge port are blocked by a valve element of the fusible plug valve when the fusible plug valve is closed, and communicate with each other when the fusible plug valve is opened; and the discharge port is provided in a part of the outer wall of the spacer other than the outer wall in contact with the valve body and other than the outer wall in contact with the fusible plug valve.
 2. The valve structure according to claim 1, wherein: the fusible plug valve includes a pressing member that presses the valve element in a valve closing direction, and a fusible member juxtaposed to the pressing member; and the discharge port is disposed on a side opposite to the fusible member when viewed in a direction orthogonal to a direction in which the pressing member and the fusible member are juxtaposed.
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